

WEEK 4

Power BI Exploratory Data Analysis

Simple Moving Averages, Volume Analytics, Interactive Slicers and Custom Tooltips

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Table of Contents

1. Introduction and Objectives
2. Dataset and Data Model
3. DAX Measures: SMA and Crossover Signal
4. Price Trend and Moving Average Chart
5. Trading Volume and Volatility Visual
6. Custom Tooltip Overlay Page
7. Tooltip Result
8. Timeline Slicers and Final Dashboard
9. Conclusion

1. Introduction and Objectives

Stock market data is time-series data: every row is a snapshot of one trading day. Raw daily values are noisy, so analysts use smoothing tools such as moving averages to reveal the underlying trend. Trading volume, the number of shares traded per day, is equally important because large volume usually confirms the strength of a price move.

Power BI is well suited to this kind of analysis because it combines data modelling, a calculation language (DAX) and interactive visuals in one tool. This week's task was to build a complete analytical dashboard on top of the Shopify stock table.

1.1 Objectives

- **SMA measures.** Calculate the 20-day and 50-day Simple Moving Average using DAX.
- **Price chart.** Build an interactive price trend chart with moving average lines and a zoom slider.
- **Volume chart.** Build a volume area chart that highlights unusual volume spikes with conditional colours.
- **Slicers.** Add date and value slicers and a dynamic metric selector for interactive filtering.
- **Tooltip.** Create a custom tooltip page that shows detailed trade information on hover.

1.2 Scope and Approach

The work was completed in the order listed above. First the calculation layer (DAX measures) was created, then the visuals were built on top of those measures, then the interactive controls were added, and finally the tooltip page was linked to the main charts. Each step is explained in the sections that follow, with screenshots taken from the Power BI Desktop report canvas.

1.3 Tools Used

Component	Purpose
Power BI Desktop	Data modelling, DAX measures and report design
DAX	Moving averages, crossover signal and conditional colour logic
Date Table (Dim_Date)	Continuous calendar for time intelligence functions
Shopify stock table	Daily open, high, low, close, adjusted close and volume

2. Dataset and Data Model

The source table is a daily Shopify (SHOP) price history file, loaded into Power BI as the fact table. Every row holds the trading data of a single day. A separate Date Table was created and related to the stock table so that time intelligence functions such as DATESINPERIOD can work on a complete, gap-free calendar.

2.1 Fact Table Columns

Column	Description
date	Trading date, the key used to relate to the Date Table
open / close	Price at market open and at market close
high / low	Highest and lowest price traded during the day
adj_close	Close price adjusted for splits and dividends
volume	Number of shares traded that day

2.2 Date Table

The Date Table provides the Date, Day, Month, Month Number, Quarter and Year fields. Using a dedicated calendar (instead of the raw date column) is a best practice because weekends and holidays, which have no stock rows, would otherwise create gaps in moving average calculations and in the continuous X-axis.

2.3 Calculated Measures

Besides the raw columns, the model contains several DAX measures that are used throughout the report:

- **Total Volume:** sum of shares traded in the selected period.
- **Volume 20 SMA / Volume 50 SMA:** rolling 20-day and 50-day average volume.
- **MA Crossover Signal:** text label showing whether the short average is above or below the long average.
- **Daily Return %:** day-over-day percentage change of the closing price.
- **Volume Color:** conditional colour code for highlighting volume spikes.
- **Price measures:** close, open, high, low and previous close price measures used in cards and tooltips.

3.4 Data Range

After loading, the date slicer shows the data running from 05-01-2017 to 14-03-2025, which is roughly eight years of daily trading history. Closing prices range from about 1.93 to 169.06,

and daily volume ranges from about 1.04 million to 208.96 million shares. This wide range is why smoothing and conditional highlighting are needed.

3. DAX Measures: SMA and Crossover Signal

A Simple Moving Average is the arithmetic mean of a metric over the last N days. As the date advances, the window slides forward, which smooths short-term noise. Both SMAs below are calculated on trading volume using DATESINPERIOD, which returns the N days ending at the current date in the filter context.

3.1 Volume 20-Day Simple Moving Average

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD( Dim_Date[Date], CurrentDate, -PeriodDays, DAY )  
RETURN  
    CALCULATE( AVERAGE(Shopify_Stock[Volume]), DateInPeriod )
```

3.2 Volume 50-Day Simple Moving Average

```
Volume 50 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD( Dim_Date[Date], CurrentDate, -PeriodDays, DAY )  
RETURN  
    CALCULATE( AVERAGE(Shopify_Stock[Volume]), DateInPeriod )
```

3.3 Moving Average Crossover Signal

```
MA Crossover Signal =  
IF( ISBLANK([20 SMA]) || ISBLANK([50 SMA]), BLANK(),  
    IF( [20 SMA] > [50 SMA],  
        "Bullish (20 > 50)", "Bearish (20 < 50)" ) )
```

3.4 How the Measures Work

1. **CurrentDate.** MAX(Dim_Date[Date]) reads the last date visible in the current filter context (for example, the date under the cursor on a chart).
2. **Window.** DATESINPERIOD returns a table of all dates from that day back over the last 20 or 50 days.
3. **Average.** CALCULATE applies that window as a filter and AVERAGE returns the mean volume inside it.
4. **Signal.** The crossover measure returns a blank when either average is missing, then compares the short and long averages: 20 above 50 is labelled bullish, otherwise bearish.

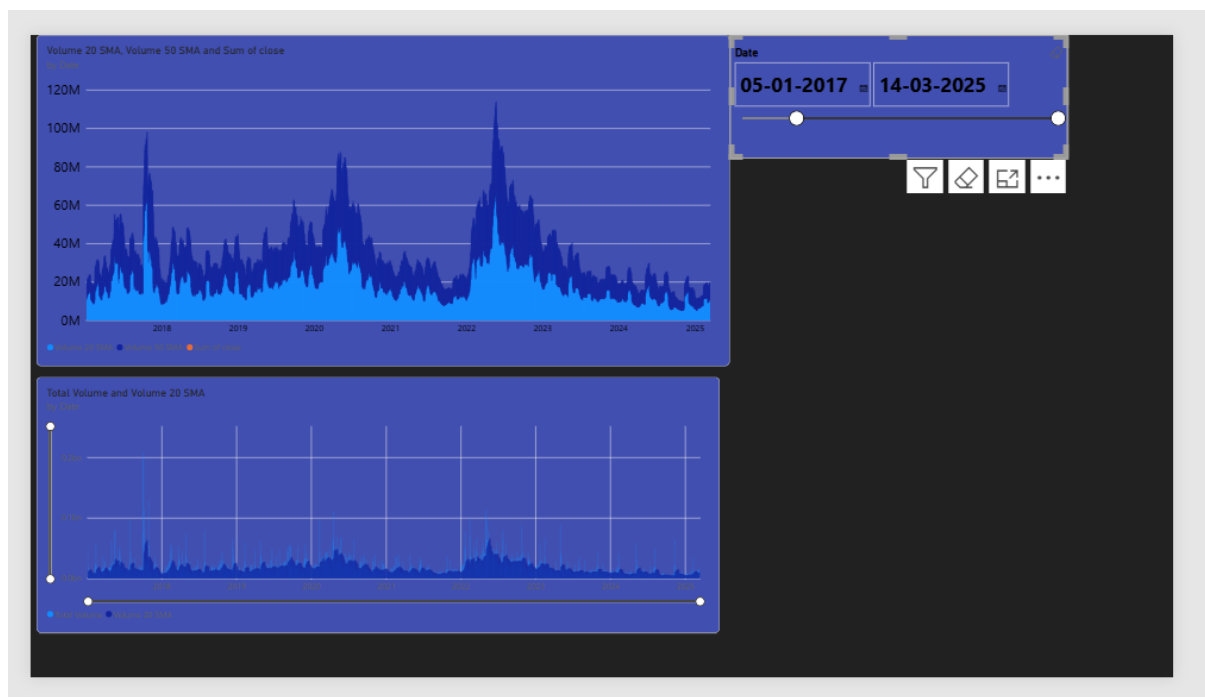
Note: the crossover formula refers to the two SMA measures; in the finished model these correspond to Volume 20 SMA and Volume 50 SMA.

4. Price Trend and Moving Average Chart

After the DAX measures were ready, the visualisation stage started. The first visual is a trend chart that places the price action and the moving averages on the same time axis so that they can be compared directly. It is built as a Line and Stacked Column chart.

4.1 Configuration Steps

1. **Visual type.** Select the Line and Stacked Column Chart from the Visualizations pane.
2. **X-axis.** Assign Date Table[Date] to the X-axis and set the axis type to Continuous so the timeline is unbroken.
3. **Y-axis.** Add Close Price, 20 SMA and 50 SMA to the line values.
4. **Zoom slider.** Under Format visual > X-axis, switch the Zoom slider On for quick timeline zooming.



It shows the report page after the two main visuals were placed. The top chart plots Volume 20 SMA, Volume 50 SMA and the close price by date, and the Date slicer in the upper right (05-01-2017 to 14-03-2025) controls both charts. Both charts share the same continuous X-axis, so the years line up vertically.

5. Trading Volume and Volatility Visual

The second visual places trading volume directly below the price chart so that volume spikes can be matched with price movements. An Area Chart (or Clustered Column chart) is used because volume is a quantity that accumulates over each day.

5.1 Configuration Steps

1. **Position.** Place the chart directly below the price chart so both share the same time scale.
2. **X-axis.** Assign Dim_Date[Date] with continuous scaling so it aligns with the top visual.
3. **Y-axis.** Add Total Volume and Volume 20 SMA.
4. **Data colour.** Apply the Volume Color measure through Format visual > Columns > fx (conditional formatting).

5.2 Conditional Colour Measure

To highlight unusual activity, a colour measure returns red when the day's volume is more than 1.5 times its 20-day average, and green otherwise:

```
Volume Color =  
IF( [Total Volume] > [20 SMA] * 1.5,  
    "#FF4D4D",    -- volume spike (red)  
    "#26A69A" )  -- normal volume (green)
```

Condition	Colour	Meaning
Total Volume > 1.5 x 20 SMA	Red (#FF4D4D)	Volume spike, unusual trading interest
Total Volume <= 1.5 x 20 SMA	Green (#26A69A)	Normal trading activity

5.3 Reading the Volume Chart

The bottom chart in Figure 1 shows that most days sit close to the 20-day average, with tall thin spikes at intervals. Those spikes are the days flagged by the Volume Color rule. Because the 20 SMA line rises after a cluster of spikes, the two series together show whether a burst of activity was a one-day event or the start of a sustained change in interest.

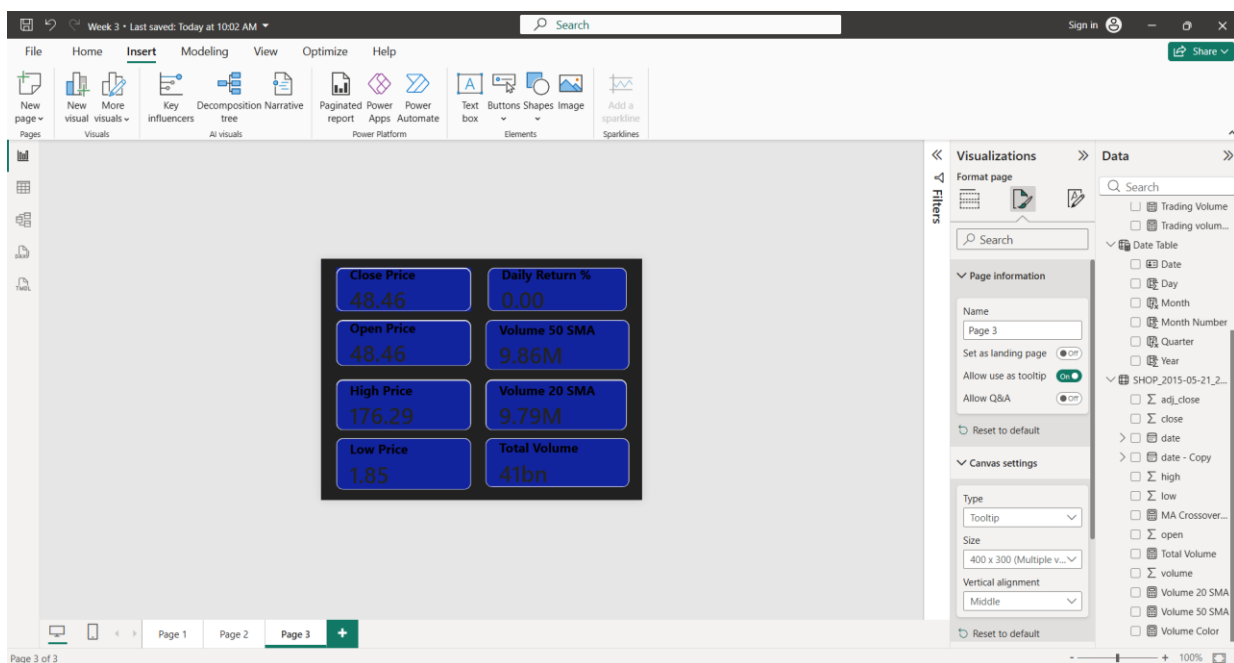
The Total Volume axis reaches into the hundreds of millions on the busiest days, which is why a continuous scale and the zoom slider are important: without zooming, the normal days appear flat next to the extreme days.

6. Custom Tooltip Overlay Page

A tooltip page is a small report page that appears when the reader hovers over a data point. It gives instant trade detail without cluttering the main dashboard. In this report the tooltip is built on Page 3.

6.1 Creating the Tooltip Page

1. **New page.** Create a new report page (Page 3 in this file; the guideline suggests naming it Stock_Tooltip).
2. **Enable tooltip.** In Format page > Page information, switch Allow use as tooltip On.
3. **Canvas type.** Under Canvas settings set Type to Tooltip. The size is 400 x 300.
4. **Add cards.** Add cards for Close Price, Daily Return %, Open Price, High Price, Low Price, Volume 20 SMA, Volume 50 SMA and Total Volume.

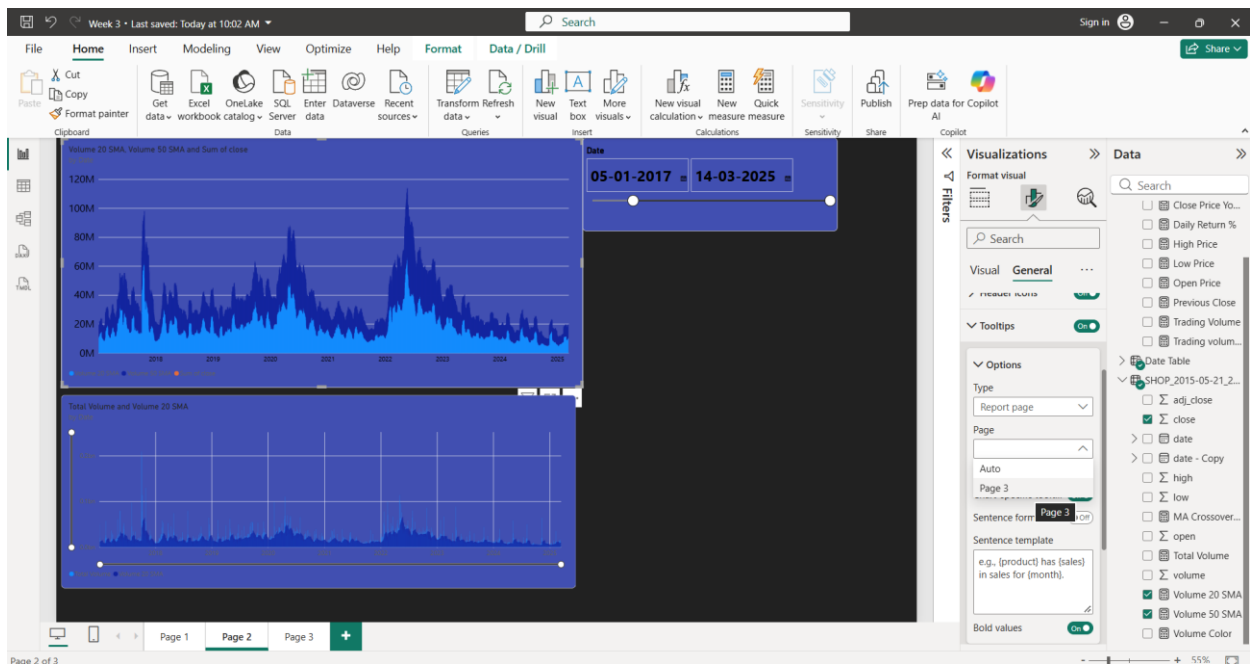


This shows the finished tooltip canvas: a dark 400 x 300 card containing eight metric cards. The values on the canvas (for example 48.46) are only design-time defaults; when the tooltip appears on a chart, each card is filtered to the date under the cursor.

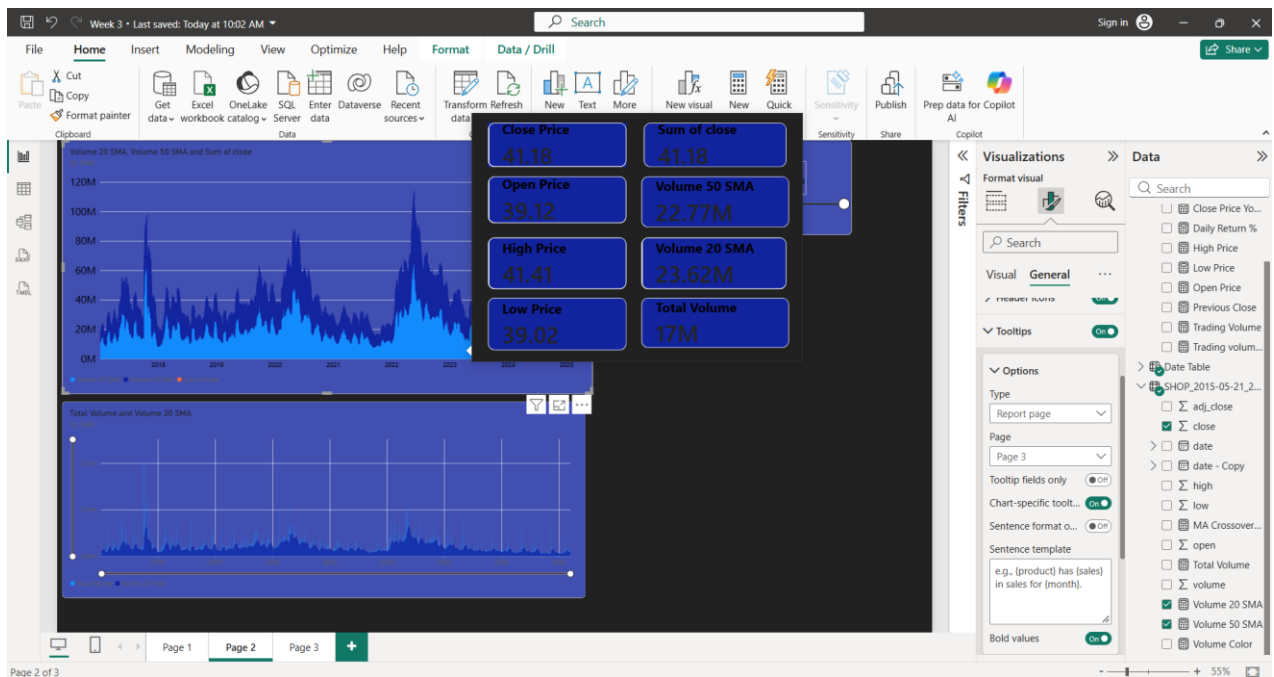
7. Tooltip Result

7.1 Linking the Tooltip to the Chart

To connect the tooltip page to a chart, select the chart and go to Format visual > General > Tooltips. Turn Tooltips On, set Type to Report page and choose Page 3 from the Page dropdown, as shown in this figure.



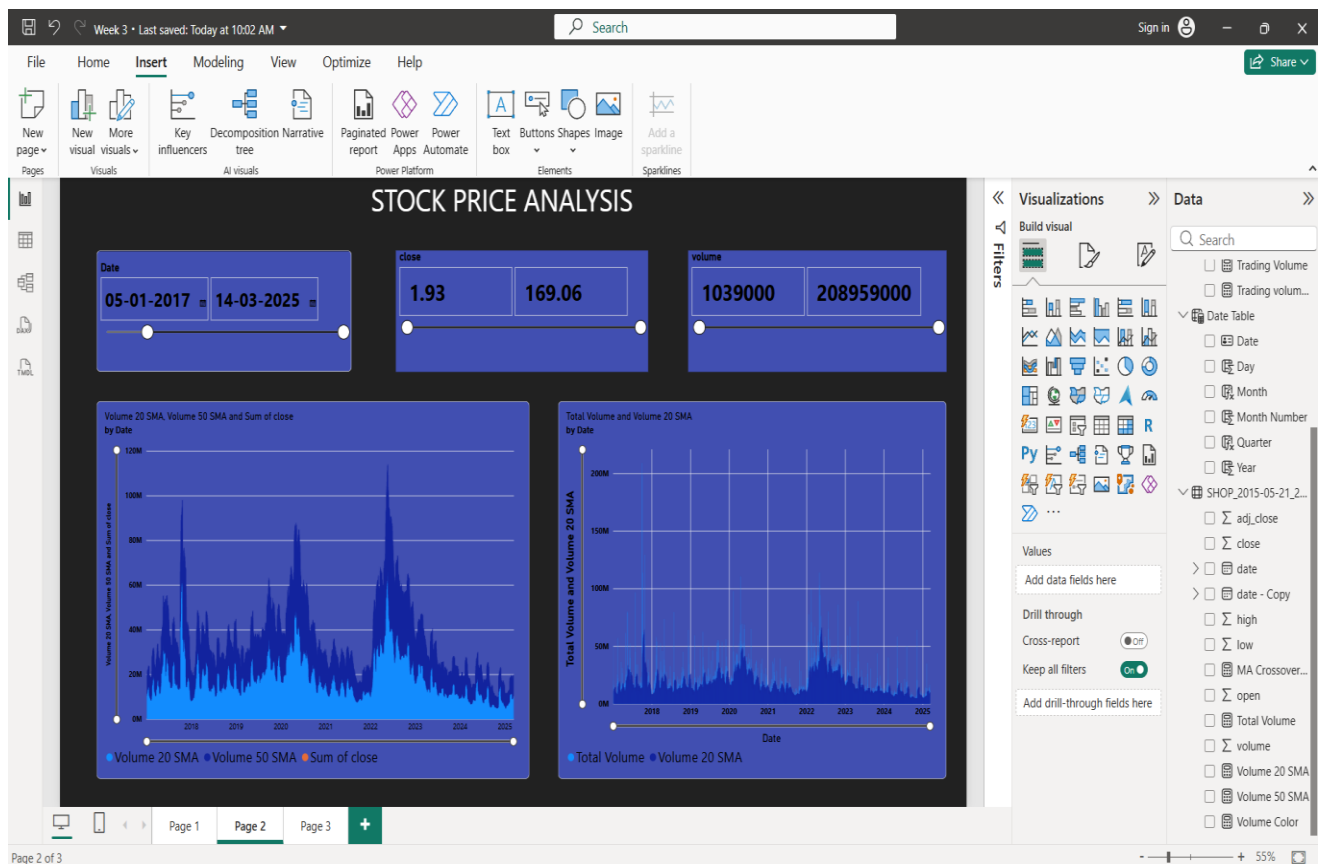
7.2 Tooltip in Action



Hovering on the chart displays the tooltip in this figure: Close 41.18, Open 39.12, High 41.41, Low 39.02, Total Volume 17M, Volume 20 SMA 23.62M and Volume 50 SMA 22.77M. Here the 20-day average is above the 50-day average, so the crossover signal would read Bullish ($20 > 50$), and the day's volume of 17M is below both averages, so no spike is flagged.

8. Timeline Slicers and Final Dashboard

Slicers let the report reader filter every visual on the page at once. The report uses three range slicers and a parameter switch, placed in a row above the charts under the title STOCK PRICE ANALYSIS.



8.1 Slicers on the Page

Slicer	Current range	Purpose
Date	05-01-2017 to 14-03-2025	Between-mode date range for both charts
close	1.93 to 169.06	Filter days by closing price range
volume	1,039,000 to 208,959,000	Filter days by number of shares traded

9 Conclusion

The Week 4 exercise produced an interactive Power BI report with DAX-driven moving averages, a crossover signal, a volume spike highlighter, range slicers and a custom hover tooltip.